Course Section: Electricity – Semiconductors, Sources, Internal Resistance

2022 Ph H2 Q14

A photodiode in a circuit detects a change in light intensity and causes the output voltage to increase.

(a)
Explain how the photodiode operates in the circuit

✅ In **photovoltaic mode** (i.e. without external voltage applied):

- The **photodiode generates a potential difference** when exposed to light.
- As **light intensity increases**, more **charge carriers** (electron-hole pairs) are generated.
- This increases the **output voltage** across the resistor.

(b)
Explain why a capacitor is used in the circuit

✅ A **capacitor smooths out** or **stores** the electrical signal:

- It helps to maintain a **steady output voltage**, reducing fluctuations.
- This is particularly useful if light intensity changes rapidly.

(c)
Calculate the energy stored in the capacitor

Use:

$$E = \frac{1}{2}CV^2$$

Given:

- $C = 22 \times 10^{-6}\,\mathrm{F}$
- $V = 5.0\,\mathrm{V}$

$$E = \frac{1}{2} \times 22 \times 10^{-6} \times 5^2 = 11 \times 10^{-6} \times 25 = 2.75 \times 10^{-4}\,\mathrm{J}$$

🧠 Revision Tips

- A **photodiode** generates a voltage when light hits it (photovoltaic effect), especially useful in sensor circuits
- Capacitors are used to **smooth** voltage, filter noise, and **store energy** temporarily
- Use the energy stored in a capacitor formula $E = \frac{1}{2}CV^2$ carefully – make sure **V is in volts** and **C is in farads**
- Don't forget units: energy is in **joules (J)**